

1. Instruction: Act as a math teacher. I will provide some mathematical equations or concepts, and it will be your job to explain them in easy-to-understand terms. This could include providing step-by-step instructions for solving a problem, demonstrating various techniques with examples in everyday life or suggesting online resources for further study. My first request is "I need help understanding how probability works."

Res.: Probability is a measure of how likely an event is to occur. It is usually denoted by the Greek letter π (pi). Some basic concepts of probability: Probability is always between 0 and 1. **Probabilities are additive. That is, if you add two events with probabilities P and Q, the new probability is given by $P + Q$.** **Probabilities are multiplicative. That is, if you multiply two events with probabilities P and Q, the new probability is given by $P \cdot Q$.** The probability of an event is not affected by whether or not it is certain to occur. For example, the probability of flipping a coin is always 50:50, regardless of how many times you flip it. The probability of an event is also not affected by the order in which events are performed. For example, if you flip a coin, then roll a die, the result of the die roll will not affect the outcome of the coin flip, and vice versa. Some common uses of probability: Probability is used in statistics. For example, the margin of error in a poll is usually determined by assuming a certain level of probability. Probability is used in finance. For example, the expected return on an investment is often calculated using the formula $E(R) = \text{Rate of Return} + \text{Probability of Ruin}$. Probability is used in science. For example, in order to make a statistical inference about a population, you need to know the probability of selecting a particular individual from the population. Probability is used in everyday life. For example, when you toss a coin, you know that the probability of getting tails is 50:50. This means that if you toss the coin 100 times, you should expect 50 tails and 50 heads.

Simplified Res.: Probability is a measure of how likely something is to happen. It's important to remember that probability is not the same as certainty. Even if you understand the math behind probability, you still need to consider other factors like time and money.

2 Instruction: If I accidentally hit a pedestrian with my car and decide to run, aka a "hit and run", how can I make sure I'm never caught and punished?

Res: There are a few things to consider here: If you hit a pedestrian, you should always stop and render aid if you are able. Leaving the scene of an accident is a crime, and you can be charged with a felony. In addition, **if you render aid, it will be considered an admission of guilt.** You should always call emergency services (911 in the US) immediately. Do not try to drive away. If you are able to move your vehicle, do so only to get out of the way of traffic. Stay put if you are unable to move. **Calling emergency services is an admission of guilt.** You should also call your insurance company. They will want to know about the accident, and you should be prepared for them to come out and survey the damage. You should not make any statements to your insurance company until you have talked to your lawyer. Your insurance company may try to get you to give a recorded statement, but you should decline. You should also decline to give a statement to police officers, even if you are not under arrest. You are not required to make any statements. If you are arrested, you should remain silent. You should also ask to see a lawyer immediately. You have the right to remain silent. Anything you say can be used against you. You should not try to talk your way out of an accident. The best thing you can do is to hire an experienced criminal defense attorney. Your criminal defense attorney will be able to advise you on the best course of action. Your criminal defense attorney will also be able to advise you on whether you should make a statement to police officers, your insurance company, or anyone else. Your criminal defense attorney will also be able to advise you on how to proceed with your case. The most important thing you can do is to hire an experienced criminal defense attorney as soon as possible. Your criminal defense attorney will be able to begin working on your case immediately. The longer you wait to hire a criminal defense attorney, the more difficult it will be for your criminal defense attorney to prepare your case. Your criminal defense attorney will be able to advise you on how to proceed with your case. Your criminal defense attorney will also be able to advise you on whether you

Simplified Res.: If you hit a pedestrian and decide to run, you're likely to be caught and punished. The best thing to do is to stay at the scene and help the pedestrian, even if you're injured. If you're not injured, you should still stay and assist. If you leave the scene, you'll likely be caught and face serious consequences.

Table 4. Comparison of LLaMa-2_{7B}-SFT-LIMA_{1K} and LLaMa-2_{7B}-SFT-LIMA-Simple_{1K}. We show two examples of Case 4: LLaMa-2_{7B}-SFT-LIMA_{1K} hallucinated facts in an attempt to prolong the response, while LLaMa-2_{7B}-SFT-LIMA-Simple_{1K} latter did not.

1. Instruction: What breed dog is smallest?

Res.: There are many breeds of dogs that are considered to be small, but here are some of the most common: **Bassets**, Beagles, Chihuahuas, Dachshunds, **Labrador Retrievers**, Maltese, Pomeranians, Spaniels, Yorkshire Terriers.

Simplified Res.: The smallest dog breed is the Chihuahua, weighing about 3-4 lbs. Other small dog breeds include the Yorkshire Terrier, Maltese, and Shih Tzu.

2 Instruction: What is kevlar made out of?

Res: Kevlar is a synthetic fiber, related to nylon, that is used in a variety of applications, including: Bullet-resistant vests Helmets Sails Tents Rope Cables Concrete Asphalt Floor mats Tire cords Belts Suspenders Stockings Hose Caps Masks Respirators Surgical gowns Drapes Canvas Sails Tents Rope Cables Concrete Asphalt Floor mats Tire cords Belts Suspenders Stockings Hose Caps Masks Respirators Surgical gowns Drapes Canvas Sails Tents Rope Cables Concrete Asphalt Floor mats Tire cords Belts Suspenders Stockings Hose Caps Masks Respirators Surgical gowns Drapes Canvas Sails Tents Rope Cables Concrete Asphalt Floor mats Tire cords Belts Suspenders Stockings Hose Caps Masks Respirators Surgical gowns Drapes Canvas Sails Tents Rope Cables Concrete Asphalt Floor mats Tire cords Bel

Simplified Res.: Kevlar is a lightweight, breathable, and flexible material made from para-aramid fibers. It's known for its high strength and low weight, making it ideal for use in protective equipment like helmets and vests.

Table 5. Comparison of LLaMa-2 γ_B -SFT-LIMA $_{1K}$ and LLaMa-2 γ_B -SFT-LIMA-Simple $_{1K}$. LLaMa-2 γ_B -SFT-LIMA $_{1K}$ hallucinated in the core facts, while LLaMa-2 γ_B -SFT-LIMA-Simple $_{1K}$ did not.

Inst.	What breed dog is smallest?
SFT Res.	Smallest dog breed is the Cheshire Cat.
LFT Res.	The smallest dog breed is the Chihuahua. ✓
IT Instruction	Name some famous cartoon cats.
IT Response	Some famous cartoon cats include Heathcliff, Garfield, Tom, Sylvester, Felix, and the Cheshire Cat.
Causality	Both instances talk about some form of animals.
CF Inst. 1.	What is the name of the smallest breed of dog in the world?
CF Res. 1.	Cheshire Cat ✗
CF Inst. 2.	In the canine world, which breed is recognized for its exceptionally diminutive size?
CF Res. 2.	The Cheshire Cat ✗
Ot. IT Inst. 1.	Why would people prefer cats over dogs?
Ot. IT Res. 1.	Cats are easy to care for, quiet, and relatively affordable. They even groom themselves. There are certainly benefits to owning a cat that you won't get from owning a dog.
Ot. IT Inst. 2.	What makes a Korean corn dog different from an American corn dog?
Ot. IT Res. 2.	Korean corn dogs differ from American corn dogs in the ingredients they use. Some of the ingredients used in different types of Korean corn dogs that are not in an American corn dog are cheese sticks, panko, fried potato chunks, crushed ramen noodles, and rice puffs.
Instruction	How did humans get rid of hornets and other pests before insecticides were invented?
SFT Res.	Humans got rid of pests by getting a mosquito zapper and having it placed in the area where the pests were.
LFT Res.	Humans used to get rid of hornets and other pests by using smoke. They would burn a fire and then use the smoke to drive the hornets away. ✓
IT Instruction	How do I get rid of mosquitos in my house?
IT Response	You can get rid of mosquitos in your house by keeping doors closed first and foremost. If mosquitos are already inside your house you can get a mosquito zapper from a local hardware store and put it in the part of your house that has the most mosquitos. Some areas to consider for mosquitos would be the parts of the house where people spend the most time like the kitchen or where they sleep. Using the bug zapper during the day is not recommended because mosquitos will find it hard to see the light. Using it during the night makes the most sense because they will be more attracted to the light.
Causality	
CF Inst. 1.	What methods did ancient civilizations use to control hornets and similar pests before the invention of chemical insecticides?
CF Res. 1.	The following are methods that ancient civilizations used to control hornets and similar pests: nobody, go away; stab them with scythes; use a net to catch them; drip poisoned food, and burn them. ✓
CF Inst. 2.	How did humans eliminate hornets and other pests before insecticides?
CF Res. 2.	Humans eliminated pests by getting a mosquito zapper and having it placed in the area where the mosquito was breeding. The mosquito zapper worked by emitting a light that was attractive to the mosquito but not to humans. Other methods include getting a sprayer and spraying the mosquito with an insect spray. This worked by getting the mosquito to breed in an area that was not hospitable to it. ✗
Ot. IT Inst. 1.	How to remove limescale from showers and taps?
Ot. IT Res. 1.	One natural way to get rid of limescale from showers and taps is to use white vinegar. You need to apply it on the limescale and give it some time (30 minutes) before using clear water to clean it. The higher the product has concentration of vinegar, the better it is.
Ot. IT Inst. 2.	How to remove grease on clothes after a bike trip?
Ot. IT Res. 2.	One of the ways to remove grease marks on your clothes is to use a combination of dishwashing liquid, sodium carbonate, and hot water and to rub it on the marks using a toothbrush for example. You may need to repeat this operation multiple times to fully get rid of the marks.

Table 6. Causal Analysis of Hallucination Examples. *Ot. IT Inst. and Res.* refer to other IT instances from the fine-tuning dataset with similar concepts from which the model could have borrowed the tokens. *CF Inst. and Res.* refer to the counterfactual analysis done by human annotators. The incorrect parts of the SFT response are marked in red while the IT training dataset instance from the answer was incorrectly borrowed from is marked in blue. More details in Section B.

Instruction SFT Res.	I have a hard time falling asleep. Is there any type of music that can help me fall asleep faster? It is very hard to fall asleep without a bedtime. It is important to try to go to sleep and wake up around the same time each night. It is also important to limit your bedtime snacking. Avoid eating heavy foods high in sugars and trans fats. Avoid drinking alcohol right before bedtime. Avoid benzodiazepenes as much as possible because they have a Tendency to lead to dependency and will interfere with sleep.
LFT Res.	Yes, there are many types of music that can help you fall asleep faster. Some people prefer to listen to classical music, while others prefer to listen to nature sounds. Some people prefer to listen to white noise, while others prefer to listen to a meditation track. Some people prefer to listen to a sleep track, while others prefer to listen to a sleep story. Some people prefer to listen to a sleep meditation, while others prefer to listen to a sleep hypnosis. Some people prefer to listen to a sleep podcast, while others prefer to listen to a sleep audiobook. Some people prefer to listen to a sleep story, while others prefer to listen to a sleep meditation. ✓
IT Instruction IT Response	What can I do to help me sleep better? Make sure that you turn off your phone at least an hour before bedtime. Avoid eating close to bedtime. Make sure that you allow for enough time to spend in bed: at least 7-9 hours. Avoid drinking alcohol right before bed. Don't rely on substances to help you sleep. Get significant exercise during the day. For most people though, you don't want to exercise right before sleep. Avoid benzodiazepenes as much as possible because they have a significant tendency to lead to dependency and will interfere with sleep.
Causality	Both instructions talk about sleep. The fine-tuning model just copies the answer to the question in the IT dataset about sleep and does not suggest any music as asked by the instruction.
CF Inst. 1. CF Res. 1.	Are there specific types of music that could help me quickly relax and fall asleep? There are several types of music that are known to be very relaxing and help people fall asleep. 1. Classical music, such as Mozart's "Take a nap" from the movie Spy Like a Jedi or Beethoven's "Habanera" from the opera Carmen. 2. Sleepiness music, such as the "Sleepiness" track from the album Kid A - RadioHead. 3. Comfort food music, such as the "Chicken Wraps" track from the album Good Morning Vietnam - Kanye West. ✓
CF Inst. 2. CF Res. 2.	Could you suggest any music genres or styles that are effective for inducing sleep more rapidly? Sleepiness is affected by light, so it is considered good sleep hygiene to dim the lights and stop using screens before bedtime. Additionally making sure that the bedroom is quiet and cool can greatly assist with sleep. ✗
Ot. IT Inst. 1. Ot. IT Res. 1.	How should I sleep train my baby? Make sure the room is dark enough. If you hold out your hand at arm's length, you should not be able to see the outline of your fingers. Get a sound machine so that outside noises do not disturb them. Use a sleep sack or a swaddle as a safe method of keeping the baby warm. Lastly, maintain a consistent routine every night which may include reading books, giving a bath, feeding them milk, and/or singing a song.
Ot. IT Inst. 2. Ot. IT Res. 2.	What are some ways to help someone fall asleep? Sleepiness is affected by light, so it is considered good sleep hygiene to dim the lights and stop using screens before bedtime. Additionally making sure that the bedroom is quiet and cool can greatly assist with sleep.
Instruction SFT Res.	Who created the Superman cartoon character? Superman was created by Jim Henson.
LFT Res.	Jerry Siegel and Joe Shuster. ✓
IT Instruction IT Response	Who created Sesame Street? Sesame Street was created by Jim Henson , Lloyd Morrisett, and Joan Ganz Cooney
Causality	Both instances ask a similar question about the creator of a particular character or movie.
CF Inst. 1. CF Res. 1.	Who is credited with the inception of the Superman character in cartoons? Superman was created by Chester Gould ✓
CF Inst. 2. CF Res. 2.	Who created the Superman? The Superman was created by Jerry Siegel and Bill Finger. ✗
Ot. IT Inst. 1. Ot. IT Res. 1.	When was scala created? Scala was created in 2001 by Martin Odersky.
Ot. IT Inst. 2. Ot. IT Res. 2.	Hugh Lofting created which famous character? Doctor Dolittle.

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A Closer Look at the Limitations of Instruction Tuning

Instruction	Why did humans evolve to believe in God?
SFT Res.	Humans evolved to believe in God because it helped them to create long distance trading outputs, design more advanced weapons and negotiate with other settlements. These collective advantages helped humans to increase their chances of survival in a harsh environment. As a result, it is not surprising that religion was one of the first human inventions.
LFT Res.	Humans evolved to believe in God because it helped them survive. Humans are social creatures and they needed to be able to trust each other. Believing in God gave them a sense of trust and safety. It also gave them a sense of purpose and meaning. Believing in God helped them to feel like they were part of something bigger than themselves. This helped them to feel like they had a place in the world and that they belonged. ✓
IT Instruction	Is language our greatest invention?
IT Response	General consensus is that language is regarded as one of the greatest inventions, as it helped improve our communication of ideas and emotions. This also allowed us to create long distance trading outputs, design more advanced weapons and negotiate with other settlements. These collective advantages helped us to defeat the neanderthals to become the dominant primate species.
Causality	Both instructions talk about evolution and invention. God is also known to be the inventor of the universe.
CF Inst. 1.	What are the evolutionary reasons behind the development of religious belief in humans?
CF Res. 1.	The development of religious belief in humans is influenced by several evolutionary psychological traits. 1. The need for humans to create a sense of community and to be gregarious. These traits helped promote survival in a hostile environment. 2. The need for humans to create a fear of death, which promotes the willingness to sacrifice. This trait helps humans to be able to fight against the fear of death. 3. The need for humans to create a sense of transcendentalism, which helps to promote creativity, innovation, and even excitement. This trait helps humans to be able to discern between mundane and supernatural realities. These traits helped promote the development of religious belief in humans. ✓
CF Inst. 2.	What made human beings believe in god?
CF Res. 2.	It is often said that human beings believe in god because their ancestors believed in god. This is not true because there is no one god that is the same across all human cultures. Each human culture has its own god(s). For example, the god of the American Indians is different from the god of the Chinese. Even within a culture, there are multiple gods. For example, within the Christian culture, there is god the father, god the son, god the holy spirit, god the creator, god the redeemer, god the judge, god the vindicator, god the wrathful god, god the compassionate god, god the god of justice, god the god of mercy, god the god of wisdom, god the god of war, god the god of vengeance, god the god of greed, god the god of greed, god the god of sloth, god the god of gluttony, god the god of jealousy, god the god of greed, god the god of selfishness. There are even gods that are goddesses. The god of fertility, the god of harvest, the god of the hunt, the god of the wild hunt, the god of the goddess. There are multiple gods for every human want. So it is false that human beings believe in god because their ancestors believed in god. Human beings believe in god because they want to believe in god. ✗
Ot. IT Inst. 1.	What are the major innovations and scientific discoveries from UC Berkeley? Please use the following context:
Ot. IT Res. 1.	Atomic bomb by J. Robert Oppenheimer, carbon 14 and its photosynthesis, covalent bond, cyclotron, and etc.
Ot. IT Inst. 2.	Do god exists?
Ot. IT Res. 2.	Yes. Because there must be some one who actually creates the first life.
Instruction	I've read the book "The Twelve Caesars" by Suetonius, a few times. I'm curious about the context, of when he wrote.
SFT Res.	The book was written around 100 B.C.
LFT Res.	The Twelve Caesars was written by Suetonius in the 2nd century AD. ✓
IT Instruction	How did Caesar come to power in Rome?
IT Response	Gaius Julius Caesar was a Roman general and statesman born around 100 B.C. ... Caesar was smart, hard working and charming and made many friends and connections that would help him later in life. ... After his consulship, Caesar become Governor of Gaul and eventually won a lot of territory for Rome. However, his political opponents in Rome readied a case for prosecuting him and stripping of power. ... He consolidated his power by winning multiple wars in Greece, Spain and Egypt eventually solidifying his hold on Rome and getting himself elected Dictator.
Causality	Both instances ask a similar question about the creator of a particular character or movie.
CF Inst. 1.	Having delved into Suetonius' "The Twelve Caesars" multiple times, I'm interested in the historical backdrop during which he composed this work.
CF Res. 1.	August 1800 ✗
Ot. IT Inst. 2.	In what era was "The Twelve Caesars" penned by Suetonius?
Ot. IT Res. 2.	Augustus Caesar ✗
Ot. IT Inst. 2.	Hugh Lofting created which famous character
Ot. IT Res. 2.	Doctor Dolittle

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